

calculate the Shannon entropy $H(L_i)$ of the final answers given by the CoT chains in each group L_i . We use a majority vote only for the K (by default, we set $K = 3$) groups with the smallest entropy. A detailed description of the algorithm is in Appendix H.

Results. We evaluate the proposed method against vanilla majority vote (i.e., self-consistency [37]) on a randomly chosen subset of 100 questions from the GPQA dataset [29], a more challenging collection of multiple-choice questions. The results in Figure 4c show that our filtered vote consistently outperforms vanilla majority vote at different sample numbers and shows little performance degradation as the sample number increases. This further underlines the importance of considering CoT length in the reasoning process.

6 Related Work

Chain-of-Thought for LLM Reasoning. CoT has become a core technique for LLMs to solve complex reasoning tasks by generating intermediate steps [38]. Numerous variants arise to enhance CoT reasoning with more structural substeps, such as least-to-most prompting [45], tree of Thoughts [42], and divide-and-conquer methods [44, 25]. These methods fundamentally treat CoT as a framework for task decomposition and subtask solving that falls in our analysis in Section 4.

Overthinking in CoT Reasoning. With the rise of powerful reasoning models like OpenAI o1, scaling test-time compute with long CoT has gained prominence [33, 7, 39, 2]. These studies often suggest that more computation like longer CoT can lead to better results. However, this is not always true. With similar interests as ours, a few concurrent works also investigated the “overthinking” phenomenon [6] where reasoning models generate excessively long CoTs for simple problems and proposed some mitigation strategies [16, 23, 24, 34]. Our analysis not only reveals the inverted-U curve of CoT length and the existence of optimal CoT length, but also provides a in-depth understanding on the scaling behaviors and simplicity bias of the optimal CoT length, as supported by both controlled experiments and theoretical analysis. This establishes a systematic explanation of overthinking and points out principled guidelines for better CoT designs.

Simplicity Bias and Occam’s Razor in Machine Learning. The simplicity bias of CoT identified in our work resonates with broader principles like Occam’s Razor, which favors simpler explanations or models. In machine learning, a ‘simplicity bias’ often refers to neural networks learning simpler functions first or being biased towards solutions with lower intrinsic complexity [1, 19]. Our findings extend this understanding to the realm of generated reasoning paths: we reveal that even structured, multi-step reasoning processes like CoT, as produced by LLMs, exhibit such simplicity bias by favoring concise reasoning paths, particularly as model capability increases.

Theoretical Understanding of CoT. Numerous studies aim to theoretically formalize the Chain-of-Thought (CoT) process and understand its effectiveness. They include analyzing CoT’s computational advantages via circuit complexity [11, 22], demonstrating how coherent reasoning paths enhance error correction and accuracy [10], and quantifying step-wise information gain from an information-theoretic standpoint [35]. Further research has shown that detailed CoT improves learning stability by affecting gradient dynamics [21], while controlled synthetic experiments have helped uncover underlying problem-solving mechanisms in LLMs [43]. Distinct from these varied theoretical explorations, our theory characterizes how CoT length influences final performance and explains its scaling behaviors through the interplay of task decomposition and error accumulation. Furthermore, our findings on CoT scaling behaviors and the consequent need for model-specific CoT structures (as discussed in Section 3.2) resonate with the concept of algorithmic alignment [41], which suggests that models perform best when the problem structure aligns with their computation structure.

7 Conclusion

In this paper, we challenged the notion that longer Chain-of-Thought (CoT) processes are invariably superior, demonstrating through extensive experiments and theoretical analysis that CoT length and accuracy typically follow an inverted U-shaped curve, implying an optimal length that balances task decomposition against error accumulation. We discovered the simplicity bias of CoT, where more capable models prefer shorter effective reasoning paths, and formally derived scaling laws for this optimal length relative to model capability and task difficulty. Practically, we showed that reinforcement learning can guide models towards this optimal CoT length, that training on

optimally-lengthed CoTs boosts performance, and proposed "Length-Filtered Vote" as a promising inference strategy. Our work underscores the critical need to calibrate CoT length, moving beyond a one-size-fits-all approach towards a principled framework where LLMs adaptively choose the right amount of thought to optimize reasoning.

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References

- [1] Devansh Arpit, Stanisław Jastrzębski, Nicolas Ballas, David Krueger, Emmanuel Bengio, Maxinder S Kanwal, Tegan Maharaj, Asja Fischer, Aaron Courville, Yoshua Bengio, et al. A closer look at memorization in deep networks. In *International conference on machine learning*, pages 233–242. PMLR, 2017.
- [2] Bradley Brown, Jordan Juravsky, Ryan Ehrlich, Ronald Clark, Quoc V. Le, Christopher Ré, and Azalia Mirhoseini. Large language monkeys: Scaling inference compute with repeated sampling, 2024. URL <https://arxiv.org/abs/2407.21787>.
- [3] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel Ziegler, Jeffrey Wu, Clemens Winter, Chris Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. Language models are few-shot learners. In H. Larochelle, M. Ranzato, R. Hadsell, M.F. Balcan, and H. Lin, editors, *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper_files/paper/2020/file/1457c0d6bfc4967418bfb8ac142f64a-Paper.pdf.
- [4] Lijie Chen, Binghui Peng, and Hongxun Wu. Theoretical limitations of multi-layer transformer, 2024. URL <https://arxiv.org/abs/2412.02975>.
- [5] Qiguang Chen, Libo Qin, Jiaqi WANG, Jingxuan Zhou, and Wanxiang Che. Unlocking the capabilities of thought: A reasoning boundary framework to quantify and optimize chain-of-thought. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024. URL <https://openreview.net/forum?id=pC44UMwy2v>.
- [6] Xingyu Chen, Jiahao Xu, Tian Liang, Zhiwei He, Jianhui Pang, Dian Yu, Linfeng Song, Qiuzhi Liu, Mengfei Zhou, Zhuosheng Zhang, Rui Wang, Zhaopeng Tu, Haitao Mi, and Dong Yu. Do not think that much for $2+3=?$ on the overthinking of o1-like llms, 2024. URL <https://arxiv.org/abs/2412.21187>.
- [7] Yanxi Chen, Xuchen Pan, Yaliang Li, Bolin Ding, and Jingren Zhou. A simple and provable scaling law for the test-time compute of large language models, 2024. URL <https://arxiv.org/abs/2411.19477>.
- [8] Yilong Chen, Junyuan Shang, Zhenyu Zhang, Yanxi Xie, Jiawei Sheng, Tingwen Liu, Shuohuan Wang, Yu Sun, Hua Wu, and Haifeng Wang. Inner thinking transformer: Leveraging dynamic depth scaling to foster adaptive internal thinking, 2025. URL <https://arxiv.org/abs/2502.13842>.
- [9] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems, 2021. URL <https://arxiv.org/abs/2110.14168>.

- [10] Yingqian Cui, Pengfei He, Xianfeng Tang, Qi He, Chen Luo, Jiliang Tang, and Yue Xing. A theoretical understanding of chain-of-thought: Coherent reasoning and error-aware demonstration, 2024. URL <https://arxiv.org/abs/2410.16540>.
- [11] Guhao Feng, Bohang Zhang, Yuntian Gu, Haotian Ye, Di He, and Liwei Wang. Towards revealing the mystery behind chain of thought: A theoretical perspective. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023. URL <https://openreview.net/forum?id=qHrADgAdYu>.
- [12] Yao Fu, Hao Peng, Ashish Sabharwal, Peter Clark, and Tushar Khot. Complexity-based prompting for multi-step reasoning, 2023. URL <https://arxiv.org/abs/2210.00720>.
- [13] Kanishk Gandhi, Ayush Chakravarthy, Anikait Singh, Nathan Lile, and Noah D. Goodman. Cognitive behaviors that enable self-improving reasoners, or, four habits of highly effective stars, 2025. URL <https://arxiv.org/abs/2503.01307>.
- [14] Jonas Geiping, Sean McLeish, Neel Jain, John Kirchenbauer, Siddharth Singh, Brian R Bartoldson, Bhavya Kailkhura, Abhinav Bhatele, and Tom Goldstein. Scaling up test-time compute with latent reasoning: A recurrent depth approach. *arXiv preprint arXiv:2502.05171*, 2025.
- [15] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- [16] Tingxu Han, Zhenting Wang, Chunrong Fang, Shiyu Zhao, Shiqing Ma, and Zhenyu Chen. Token-budget-aware llm reasoning. *arXiv preprint arXiv:2412.18547*, 2024.
- [17] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. In *International Conference on Learning Representations*, 2021. URL <https://openreview.net/forum?id=d7KBjmI3GmQ>.
- [18] Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the math dataset, 2021. URL <https://arxiv.org/abs/2103.03874>.
- [19] Minyoung Huh, Hossein Mobahi, Richard Zhang, Brian Cheung, Pulkit Agrawal, and Phillip Isola. The low-rank simplicity bias in deep networks. *arXiv preprint arXiv:2103.10427*, 2021.
- [20] Mingyu Jin, Qinkai Yu, Dong Shu, Haiyan Zhao, Wenyue Hua, Yanda Meng, Yongfeng Zhang, and Mengnan Du. The impact of reasoning step length on large language models. In *Annual Meeting of the Association for Computational Linguistics*, 2024. URL <https://api.semanticscholar.org/CorpusID:266902900>.
- [21] Ming Li, Yanhong Li, and Tianyi Zhou. What happened in llms layers when trained for fast vs. slow thinking: A gradient perspective, 2024. URL <https://arxiv.org/abs/2410.23743>.
- [22] Zhiyuan Li, Hong Liu, Denny Zhou, and Tengyu Ma. Chain of thought empowers transformers to solve inherently serial problems, 2024. URL <https://arxiv.org/abs/2402.12875>.
- [23] Haotian Luo, Li Shen, Haiying He, Yibo Wang, Shiwei Liu, Wei Li, Naiqiang Tan, Xiaochun Cao, and Dacheng Tao. O1-pruner: Length-harmonizing fine-tuning for o1-like reasoning pruning. *arXiv preprint arXiv:2501.12570*, 2025.
- [24] Xinyin Ma, Guangnian Wan, Runpeng Yu, Gongfan Fang, and Xinchao Wang. Cot-valve: Length-compressible chain-of-thought tuning. *arXiv preprint arXiv:2502.09601*, 2025.
- [25] Zijie Meng, Yan Zhang, Zhaopeng Feng, and Zuozhu Liu. Dcr: Divide-and-conquer reasoning for multi-choice question answering with llms, 2024. URL <https://arxiv.org/abs/2401.05190>.
- [26] Sania Nayab, Giulio Rossolini, Giorgio Buttazzo, Nicolamaria Manes, and Fabrizio Giacomelli. Concise thoughts: Impact of output length on llm reasoning and cost, 2024. URL <https://arxiv.org/abs/2407.19825>.

- [27] Qwen, :, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tianyi Tang, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. Qwen2.5 technical report, 2025. URL <https://arxiv.org/abs/2412.15115>.
- [28] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, Ilya Sutskever, et al. Language models are unsupervised multitask learners. *OpenAI blog*, 1(8):9, 2019.
- [29] David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R. Bowman. Gpqa: A graduate-level google-proof q&a benchmark, 2023. URL <https://arxiv.org/abs/2311.12022>.
- [30] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms, 2017. URL <https://arxiv.org/abs/1707.06347>.
- [31] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models, 2024. URL <https://arxiv.org/abs/2402.03300>.
- [32] Guangming Sheng, Chi Zhang, Zilingfeng Ye, Xibin Wu, Wang Zhang, Ru Zhang, Yanghua Peng, Haibin Lin, and Chuan Wu. Hybridflow: A flexible and efficient rlhf framework. In *Proceedings of the Twentieth European Conference on Computer Systems*, EuroSys '25, page 1279–1297. ACM, March 2025. doi: 10.1145/3689031.3696075. URL <http://dx.doi.org/10.1145/3689031.3696075>.
- [33] Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling llm test-time compute optimally can be more effective than scaling model parameters, 2024. URL <https://arxiv.org/abs/2408.03314>.
- [34] Yang Sui, Yu-Neng Chuang, Guanchu Wang, Jiamu Zhang, Tianyi Zhang, Jiayi Yuan, Hongyi Liu, Andrew Wen, Shaochen Zhong, Hanjie Chen, et al. Stop overthinking: A survey on efficient reasoning for large language models. *arXiv preprint arXiv:2503.16419*, 2025.
- [35] Jean-Francois Ton, Muhammad Faaiz Taufiq, and Yang Liu. Understanding chain-of-thought in llms through information theory, 2024. URL <https://arxiv.org/abs/2411.11984>.
- [36] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023.
- [37] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *The Eleventh International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=1PL1NIMMrw>.
- [38] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, brian ichter, Fei Xia, Ed Chi, Quoc V Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, editors, *Advances in Neural Information Processing Systems*, volume 35, pages 24824–24837. Curran Associates, Inc., 2022.
- [39] Yangzhen Wu, Zhiqing Sun, Shanda Li, Sean Welleck, and Yiming Yang. Scaling inference computation: Compute-optimal inference for problem-solving with language models. In *The 4th Workshop on Mathematical Reasoning and AI at NeurIPS'24*, 2024. URL <https://openreview.net/forum?id=j7DZWSc8qu>.
- [40] Yunhui Xia, Wei Shen, Yan Wang, Jason Klein Liu, Huifeng Sun, Siyue Wu, Jian Hu, and Xiaolong Xu. Leetcodedataset: A temporal dataset for robust evaluation and efficient training of code llms, 2025. URL <https://arxiv.org/abs/2504.14655>.